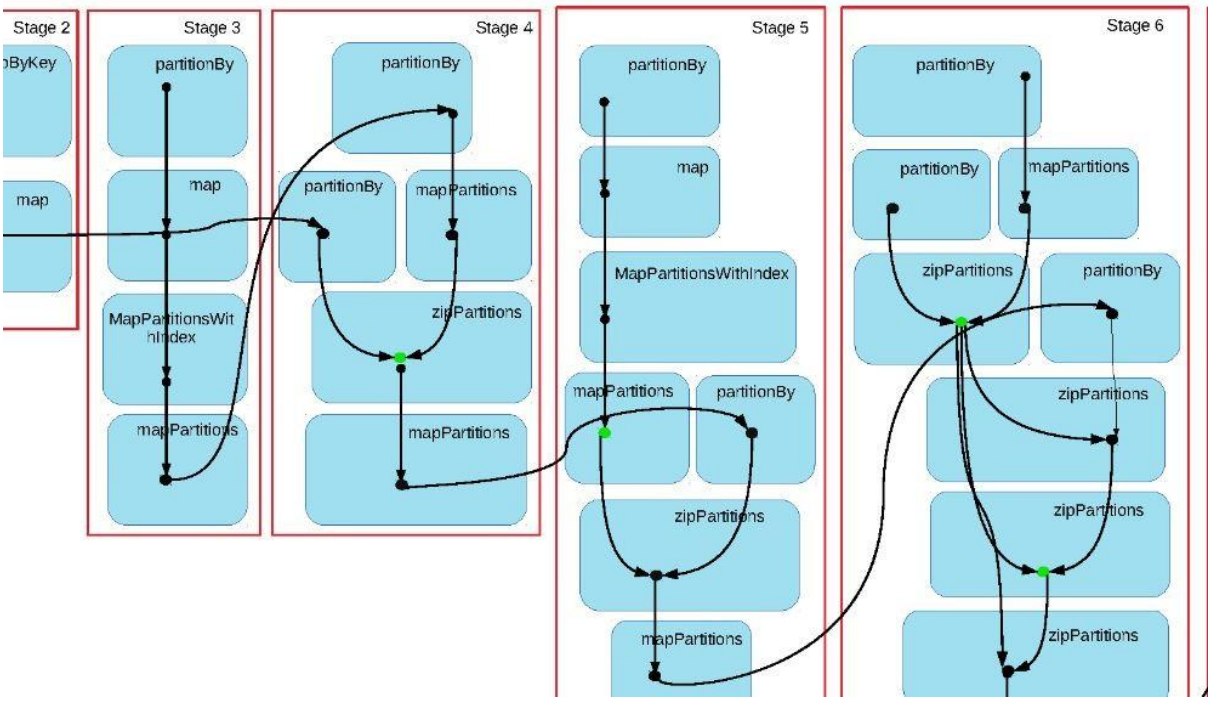
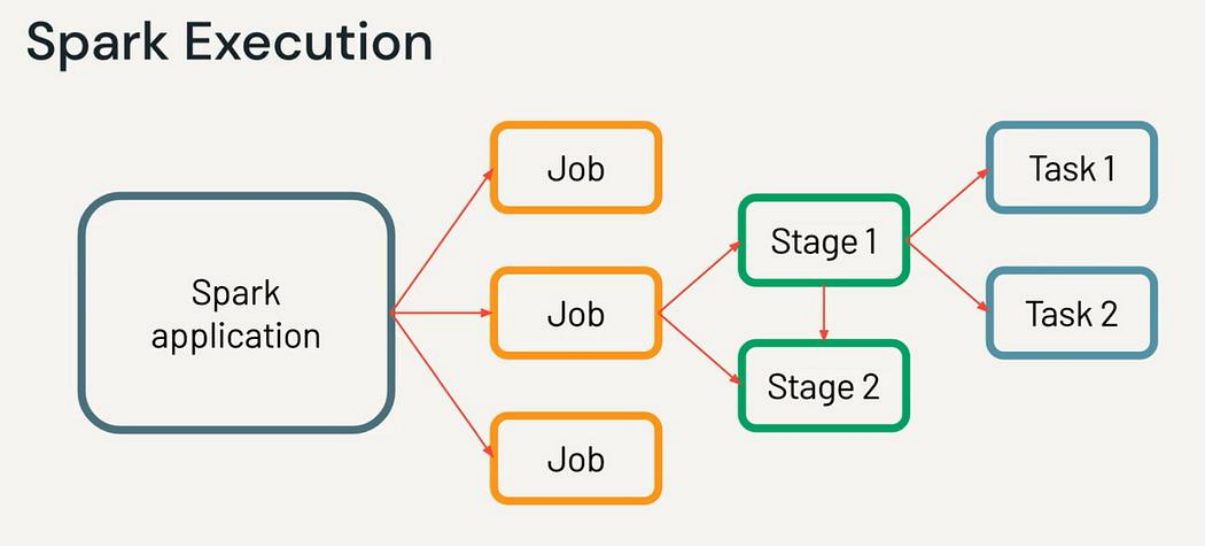
DAG

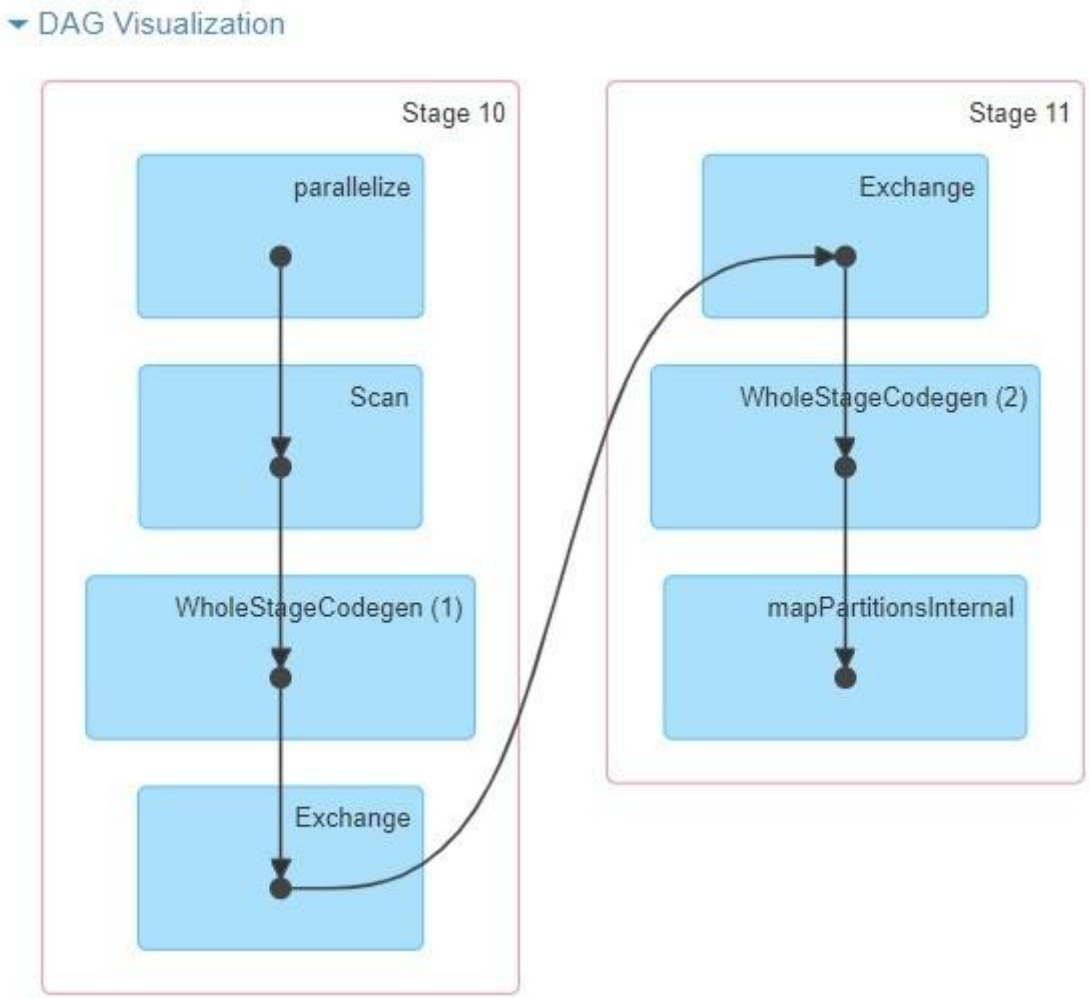
When you write Spark code using a SparkSession and apply transformations, Spark automatically builds a DAG (Directed Acyclic Graph).

You **do not create a DAG explicitly** — Spark creates it **for you**.

Let's break it down step by step (VERY IMPORTANT)







1. What you write vs what Spark builds

What YOU write (Spark code)

```
spark = SparkSession.builder.getOrCreate()

df = spark.read.csv("cdr.csv", header=True)

filtered = df.filter(df.call_type == "INTL")

grouped = filtered.groupBy("region").count()

grouped.show()
```

What SPARK does internally

Spark converts this code into a **DAG** like:

Read CSV
↓
Filter (INTL calls)
↓
GroupBy (region)
↓
Count
↓
Show (action)

This flow is the **DAG**

2/ Why Spark needs a DAG

Before Spark existed, systems worked like this:

Step 1 runs → stores output
Step 2 reads output → stores output
Step 3 reads again

The challenge was : Very slow, disk-heavy, no optimization

Spark's idea

"Let me **see the full plan first**, then I'll decide the best way to execute it."

That **execution plan = DAG**

3. What exactly creates the DAG?

Important clarification

- **SparkSession** → *entry point*
- **Transformations** → *build the DAG*
- **Actions** → *trigger the DAG execution*

Example

df.filter(...).select(...).groupBy(...).count()

DAG is **created (** Nothing runs yet)

.show()

DAG is **executed**

4. Transformations vs Actions (DAG perspective)

Transformations → DAG BUILDERS

Operation	**Effect**
filter()	Adds a node
select()	Adds a node
groupBy()	Adds a node
join()	Adds a node

Spark keeps **adding nodes to the DAG**

Actions → DAG TRIGGERS

Action	**What happens**
show()	DAG executes
count()	DAG executes

Action	What happens
collect()	DAG executes
write()	DAG executes

5 Is DAG created for every Spark program?

YES — whenever:

- You use **RDD / DataFrame / Dataset**
- You apply **transformations**
- You call **an action**

NO DAG if:

print("Hello")

(no Spark involved)

6 Why DAG is powerful

Use these in your class

Impact statements

- "Spark does **nothing** until it sees an action"
- "Transformations are just **promises**, DAG is the **contract**"
- "DAG lets Spark **optimize before execution**"
- "No DAG → No performance → No Spark advantage"

7 Real-world telecom example

Business flow

CDR file
→ Filter failed calls
→ Join SLA reference
→ Identify breaches
→ Aggregate by region
→ Write parquet

Spark builds a DAG so it can:

- Push filters early
- Reduce shuffle

- Optimize joins
- Minimize data movement

8. Final crystal-clear conclusion

Statement refined:

When we write Spark code using a SparkSession and apply transformations, Spark builds a DAG internally. The DAG is executed only when I call an action.